

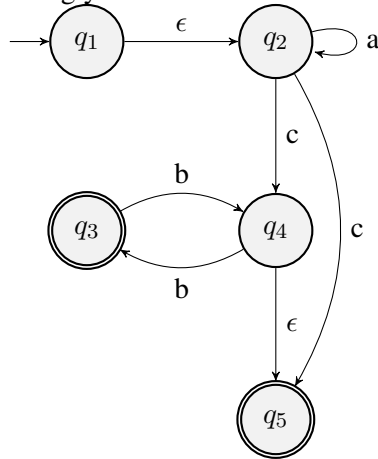
CS410 Midterm 1 Part (a)

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I. MULTIPLE CHOICE QUESTIONS (10 POINTS)

Given the following NFA M_1 . Answer questions 1-3 accordingly.



1. Which of the following can be the input alphabet Σ ?

- a) $\{a, b\}$
- b) $\{b, c\}$
- c) $\{a, b, c\}$
- d) $\{a, b, c, \epsilon\}$

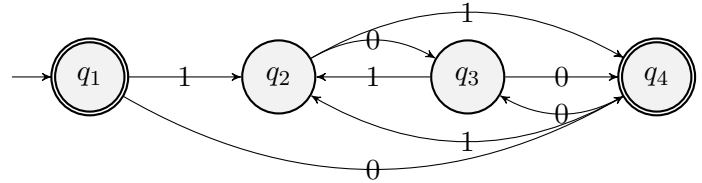
2. What is $\delta(q_2, c)$?

- a) $q_5, \{q_4\}$
- b) $q_4, \{q_5\}$
- c) q_4, q_5
- d) $\{q_4, q_5\}$
- e) q_4 or q_5

3. How many string of length less than 5 are accepted by M_1 ?

- a) 5
- b) 7
- c) 8
- d) 9
- e) 10

Given the following DFA M_2 . Answer the question 4 accordingly.



4. Which states will be not traversed for any input as the second state, third state?

- a) q_2, q_1
- b) q_2, q_3
- c) q_3, q_4
- d) q_3, q_1
- e) q_1, q_3

5. If there are 7 states in an NFA and 2 of them are final states, how many final states will the converted DFA have at maximum?

- a) 21
- b) 56
- c) 128
- d) 64
- e) 96

6. Given a nonempty regular language A (without epsilon or empty set as a member) on symbol set $\Sigma = \{0, 1\}$, how many words are of length less than 5 in minimum and maximum in A^* ?

- a) 5, 20
- b) 5, 31
- c) 20, 120
- d) 32, 1024
- e) 4, 15

7. If A and B are regular, which of the following language is not regular?

- a) $A \cup B$
- b) $A \cap B$
- c) $A \setminus B$
- d) $\overline{A}$
- e) All of them are regular

8. What is the result of the regular expressions $\emptyset^*$, RR^* ?

- a) $\emptyset$, R^+
- b) $\{\epsilon\}$, R^+
- c) $\{\epsilon\}$, R^*
- d) $\{\emptyset, \epsilon\}$, R^*
- e) $\{\emptyset\}$, R^+

9. Given the language $A = \{1^*0^*\}$, and p the pumping length. Let say we wanted to prove that A is not regular although it is regular. Let say the selected string is $1^{p/2}0^{p/2}$. Which of the following can be string y , that can not be pumped?

- a) $0^{p/2}$
- b) $1^{p/2}$
- c) 1100
- d) 0000
- e) None of the above

10. Given the language $A = \{0^i1^j | i > j\}$, and p the pumping length. Let say we wanted to prove that A is not regular. Which of the following string would you choose and pump down or up to prove it?

- a) 0^p1^{p+1} , pump down
- b) 0^p1^{p+1} , pump up
- c) $0^{p+1}1^p$, pump up
- d) $0^{p+1}1^p$, pump down
- e) $0^{p+1}1^p$, both